MTH240

Specimen Exam Paper

Engineering Mathematics I

Time allowed: 2 hours

INSTRUCTIONS TO STUDENTS:

1. This examination contains **FIVE (5)** questions and comprises **THREE (3)** printed pages (including cover page).

2. You must answer **ALL** questions.

3. All answers must be written in the answer book. Marks will only be awarded if FULL working is shown.

4. This is **an Open-book** examination.

At the end of the examination
Please ensure that you have written your examination number on each answer book used.

Failure to do so will mean that your work cannot be identified.
If you have used more than one answer book, please tie them together with the string provided.

THE UNIVERSITY RESERVES THE RIGHT NOT TO MARK YOUR SCRIPT IF YOU FAIL TO FOLLOW THESE INSTRUCTIONS.

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Question 1

A vector v is said to be a linear combination of the vectors $v_1, v_2, \dots, v_n$ if there exist scalars $c_1, c_2, \dots, c_n$ such that:

$$\boldsymbol{v} = c_1\boldsymbol{v_1} + c_2\boldsymbol{v_2} + \ \dots + c_n\boldsymbol{v_n}$$

Set up a linear system and use Gaussian elimination method to determine whether $[1,0,-1,1]^T$ is a linear combination of $[-1,1,2,0]^T$, $[2,1,0,1]^T$ and $[0,1,0,-1]^T$.

(20 marks)

Question 2

Find the eigenvalues and eigenvectors of matrix M below. Is the matrix diagonalizable?

$$M = \begin{pmatrix} 2 & 0 & 1 \\ -1 & 2 & 3 \\ 1 & 0 & 2 \end{pmatrix}$$

(25 marks)

Question 3

(a) Show that the below vector space is a subspace of $\mathbb{R}^3$:

$$V = \left\{ \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ in } \mathbb{R}^3 \mid x - y + 2z = 0 \right\}$$

Then find its basis.

(10 marks)

(b) Find a basis for the row and column space of matrix A:

$$A = \begin{pmatrix} 1 & 0 & 2 \\ -1 & 3 & 1 \\ 2 & 3 & 5 \end{pmatrix}$$

(5 marks)

Question 4

A radioactive source emits particles following a Poisson distribution, with an average rate of 4 particles per 5-second period.

(a) Determine the probability that exactly 3 particles are emitted during a 5-second period.

(7 marks)

(b) Determine the probability that at least one particle is emitted during a 5-second period.

(4 marks)

(c) For a 20-second period, what is the probability that exactly 10 particles are emitted?

(7 marks)

(d) Suppose 8 independent 5-second intervals are observed, what is the probability that exactly 5 of these intervals have exactly 3 particles emitted?

(7 marks)

Question 5

The battery life of a smartphone follows a normal distribution with a mean of 6.5 hours and a standard deviation of 2.3 hours.

(a) What proportion of smartphones last less than 5 hours on a single charge?

(5 marks)

(b) What is the probability that the battery life of a smartphone is between 4 and 10 hours?

(5 marks)

(c) How many hours of battery life will it take for 35% of smartphones to run out of charge?

(5 marks)

----- END OF PAPER -----